

Programme	B. Sc. Mathematics Honours			
Course Code	MAT2MN106			
Course Title	OPTIMIZATION TECHNIQUES IN ECONOMICS			
Type of Course	Minor			
Semester	II			
Academic Level	100 - 199			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Higher Secondary Mathematics			
Course Summary	This course examines the causes, effects, and measures of income inequality, including its measurement using tools like the Lorenz curve and Gini ratio. It explores calculus of several variables, focusing on directional derivatives, gradients, and optimization techniques, both constrained and unconstrained, with applications in economic contexts such as profit maximization and monopolistic practices. Additionally, the course covers input-output analysis, introducing technological coefficient matrices and models to analyse economic equilibrium and production functions.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Analyse the causes and effects of income inequality and evaluate the measures used to reduce it.	An	C	Internal Exam/ Assignment/ Seminar/ Viva/ End Sem Exam
CO2	Apply the principles of calculus to optimize economic functions without constraints.	Ap	P	Internal Exam/ Assignment/ Seminar/ Viva/ End Sem Exam
CO3	Evaluate constrained optimization problems using appropriate mathematical techniques.	E	P	Internal Exam/ Assignment/ Seminar/ Viva/ End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text book:		1 Micro Economic Theory(6/e), M.L.Jhingan, Vrinda publications. 2. Mathematics for Economists, Carl.P.Simon, Lawrence Blume, W.W. Norton& Company, Inc(1994) ISBN 0-393-95733-0. 3. Mathematics for Economics(Revised Edn), Mehta- Madnani, S. Chand.					
Module	Unit	Content	Hrs (48 +12)	Ext. Marks (70)			
I	Inequalities in Income -Text (1) (Chapter 47)		10	Min 15			
	1	Inequalities in Income- Causes of inequality					
	2	Effects of inequality – measures to reduce inequality					
	3	Measurement of inequality of income- Lorenz curve Gini ratio					
II	Calculus of Several Variables and Unconstrained Optimization Text(2)(Chap 14: 14.6,14.7,14.8, Chap 17: sec.17.1 to 17.5)		14	Min 15			
	4	Directional derivatives and gradients, the gradient vector					
	5	Approximation by differential Jacobian derivative					
	6	The chain rule, higher order derivative					
	7	Second order derivatives and Hessians					
	8	Young’s theorem, economical applications					
	9	Unconstrained optimization: definitions, first order conditions, second order conditions					
	10	Global maxima and minima, global maxima of concave functions					
	11	Economic applications- profit maximising firm- discriminating Monopolist					
	12	Least square analysis					
	III	Constrained Optimization - Text (2) (Chap 18: sec.18.1 to 18.7)			12	Min 15	
		13			First order conditions: objective function, constraint functions, examples		
14		Equality constraints, two variables and one equality constraints, several equality constraints					
15		Inequality constraints, one inequality constraint, several inequality constraints					

	16	Mixed constraints, constrained minimization problems		
	17	Kuhn-Tucker formulation, examples and applications		
IV	Input output analysis - Text (3) (Chap 19 :sec.19.1 to19.7,19.9,19.11,19.13)		12	
	18	Introduction- assumption- technological coefficient matrix		Min 15
	19	Closed and open input output model- coefficient matrix and open model		
	20	The Hawkins- Simon conditions- solution for two industries		
	21	Determination of equilibrium of prices- coefficient matrix and closed model		
	22	The Leontief production function- limitation of input output analysis		
V	Open Ended Module		12	
	The total derivative, The chain rule, Level curves and their tangents, Concave and Convex Functions			
References: 1. Mathematical Analysis for Economists, R G D Allen, Macmillan. 2. Fundamentals of Mathematical Economics(4/e), A C Chiang& K Wainwright, McGraw Hill. 3. Mathematical Optimization and Economic Theory (Classics in Applied Mathematics), Michael D Intriligator, SIAM(2002)				

Note: 1) Proofs of all the results are exempted for external exam. (2) 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.

Mapping of COs with PSOs and POs :

	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	2	3	2	2	1	3	2	1
CO 2	3	2	3	1	2	1	3	1	1
CO 3	2	2	3	1	2	1	3	1	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓